

MAT122: Introductory Mathematics II

Comprehensive Tutorial: Week 5 - Differentiation

Department of Mathematics

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1 Natural Logarithmic Functions

1.1 Basic and Chain Rule Applications

Find the derivative of the following functions:

1. $f(x) = 7 \ln(x^3 - 2x - 1)$

2. $g(x) = \ln(\cos x)^2$

3. $h(x) = \frac{\ln(2e^x + 1)}{3}$

4. $y = (\ln x)^4$

5. $y = \ln(\ln x^2)$

1.2 Product and Quotient Rules Involving Logarithmic functions

Find $\frac{dy}{dx}$ or the indicated derivative:

1. $y = 3x^2 \ln(\tan x)$

2. $g(t) = \frac{\ln(t^{\frac{2}{3}} - t^{-3})}{6t^2}$

1.3 Using Logarithmic Properties As Aids

Expand the expressions using logarithmic properties before differentiating:

1. $y = \ln(\sqrt{x^2 - 1})$

2. $y = \ln[2e^{-t} \sin t(t^2 + 3)]^3$

3. $y = \ln \sqrt[3]{\frac{x-1}{(3x^2+1)^4}}$

4. $f(x) = \ln \left(\frac{\sqrt{4+x^2}}{7x} \right)$

1.4 Logarithmic Differentiation

Differentiate the following: Use logarithmic differentiation to find y' :

1. $y = (x^2 + 1)^4(2x - 3)^3$

2. $y = \frac{\sqrt{x^9} \cos x}{(5-x^8)^3}$

3. $y = \frac{\sin 2x \sqrt{(x^3 + 2x + 1)^5}}{(x - 1)^2}$

4. $y = \sqrt[3]{\frac{e^{-x}-1}{16x \cos x(x+1)^7}}$

Section 2: Differentiation of Exponential Functions

1.5 Exponential Functions and Other Bases Derivatives

1. $f(x) = 8e^{-3+x}$

2. $f(x) = 3e^{\tan x}$

3. $f(x) = (\frac{1}{3})^{2x}$

4. $f(x) = \pi^{\ln x}$

5. $f(x) = \log_5(3x + 2)$

6. $f(x) = \log_4\left(\frac{\sqrt{x}}{e^x + 1}\right)$

Section 3: Implicit Differentiation

1. Find $\frac{dy}{dx}$ given $y^3 - 5y - x^3 = 9y^2 - 4$

2. Find $\frac{dy}{dx}$ for $\sqrt{xy} - \cos y = x$

3. Given $y - 5xy - x^3 = -4$ find $\frac{dy}{dx}$

4. Find $\frac{dy}{dx}$ for $e^{xy} + 4x = y^2 - x$

5. Find the equation of the tangent line to $x^2(x^4 + y^3) = y^2$ at $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$

6. Calculate the second derivative $\frac{d^2y}{dx^2}$ for $\frac{x^2}{5} + \frac{7}{y^2} = 25$